$$k = \left(\frac{2.303}{t}\right)\log\left(\frac{[A_0]}{[A]}\right)$$

$[A_0] = 100\ ;\ [A] = 25$

$$6.909 = \left(\frac{2.303}{t}\right)\log\left(\frac{100}{25}\right)$$

$$t = \left(\frac{2.303}{6.909}\right)\log(4)$$

$$t = \left(\frac{1}{3}\right)\log 2^2$$

$$t = \left(\frac{2}{3}\right)\log 2$$

14. வாய்ப்பு(இ)

$$k = \left(\frac{1}{t}\right)\ln\left(\frac{[A_0]}{[A]}\right)$$

$[A_0] = 0.1\ ;\ [A] = 0.05$

$$k = \left(\frac{1}{t_{1/2}}\right)\ln\left(\frac{0.1}{0.05}\right)$$

$$k = \left(\frac{1}{t_{1/2}}\right)\ln(2)$$

$$t_{1/2} = \frac{\ln(2)}{k}$$

15. வாய்ப்பு(ஆ)

$\text{rate}_1 = k[0.1]^n[0.1]^m$ ———(1)

$\text{rate}_2 = k[0.2]^n[0.1]^m$ ———(2)

$(2) \big/ (1)$

$$\frac{2x}{x} = \frac{k[0.2]^n[0.1]^m}{k[0.1]^n[0.1]^m}$$

$$\frac{2x}{x} = 2^n \quad \therefore\ n = 1$$

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